Observe that, defining

$$\boldsymbol{G} = \begin{bmatrix} 0 \\ \boldsymbol{p} \end{bmatrix}$$

yields the kinetic temperature observable $k_{\mathrm{B}}\mathrm{T} = \mathrm{Av}_{\beta}(\boldsymbol{p}^T \boldsymbol{M}^{-1}\boldsymbol{p})/\mathrm{N}_{\mathrm{c}}$, whereas

$$\boldsymbol{G} = \begin{bmatrix} \nabla U(\boldsymbol{q}) \\ 0 \end{bmatrix}$$

gives the *configurational* temperature observable (6.5).

An even simpler configuration-dependent choice is derived from

$$\boldsymbol{G} = \begin{bmatrix} \boldsymbol{q} \\ 0 \end{bmatrix},$$

corresponding to

$$k_{\mathrm{B}}\mathrm{T} = \frac{\mathrm{Av}_{\beta}(\boldsymbol{q} \cdot \nabla U)}{\mathrm{N}_{\mathrm{c}}}.$$

An important point is that the formula of Rugh may not be robust from a computational perspective in cases where the denominator may be close to zero. The simple choice of the kinetic temperature is often desirable since it works for both the infinite domain and for periodic boundary conditions and is such that the denominator is a nonzero constant.

Rugh's principle gives generalized temperature indicators on an infinite domain, but as we have already pointed out, typical molecular dynamics models are confined by periodic boundary conditions to a torus in configuration space (or a toric-cylinder in phase space). Similar results may be obtained in this case. The key point in the proof above is the vanishing contribution from the boundary as $R \to \infty$; in the case of periodic boundary conditions, the boundary terms from opposite sides of the domain can be seen to cancel out, leaving the desired relation. In general one should ensure that the formula for temperature is invariant under translations of the periodic domain along coordinate directions in the configurational domain. If $\boldsymbol{G}$ and H are both periodic in $\boldsymbol{q}$ and $\boldsymbol{G}$ is smooth, then everything works since the outward pointing normal will be in opposite directions on each side of the simulation cell, and the boundary term will cancel.

Thus for periodic boundary conditions we may choose

$$\boldsymbol{G}(z) = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{p} \end{bmatrix},$$